



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 10/08/2026

EXPERIMENT 18-- DATABASE WITH AUTO_INCREMENT SEQUENCES

AIM:

To implement an **Explicit Cursor in MySQL** to retrieve student names from the student_cgpa table **row by row** and build a list of students.

PROCEDURE:

1. Open MySQL Workbench and select the database.
2. Check the student_cgpa table.
3. Declare an explicit cursor to select student names.
4. Open the cursor and fetch records one by one.
5. Store each student name in a list.
6. Close the cursor after all records are fetched.
7. Call the procedure and display the result.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
  reg_no VARCHAR(15) PRIMARY KEY,  
  student_name VARCHAR(50) NOT NULL,  
  department VARCHAR(20),  
  no_of_subs_pass INT,  
  no_of_subs_fail INT,  
  cgpa DECIMAL(3,2)  
);
```

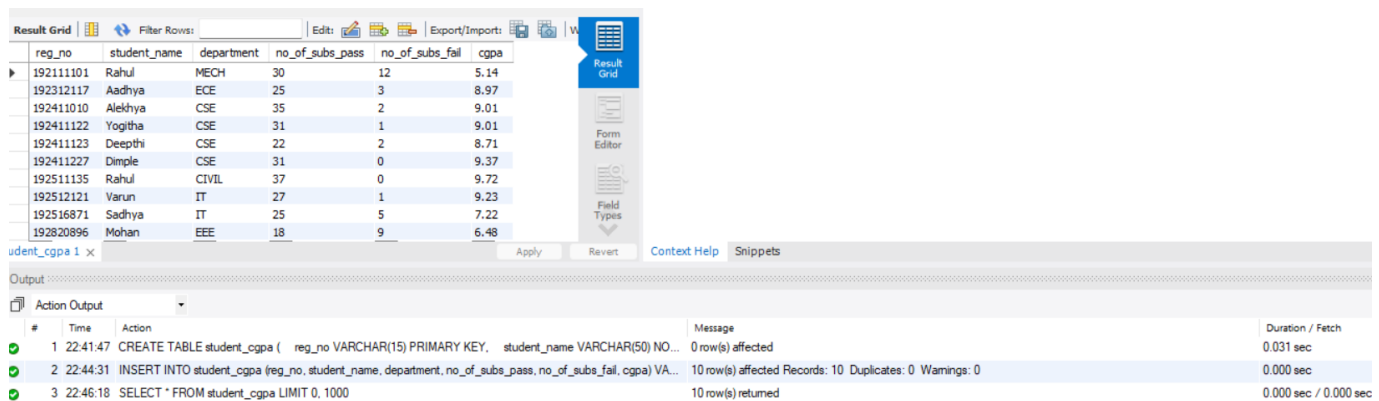
Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123, 'Deepthi', 'CSE', 22, 2, 8.71),  
(192512121, 'Varun', 'IT', 27, 1, 9.23),  
(192312117, 'Aadhya', 'ECE', 25, 3, 8.97),  
(192411227, 'Dimple', 'CSE', 31, 0, 9.37),
```

```
(192820896,'Mohan','EEE',18,9,6.48),
(192411122,'Yogitha','CSE',31,1,9.01),
('192516871','Sadhya','IT',25,5,7.22),
(192111101,'Rahul','MECH',30,12,5.14),
(192411010,'Alekhyia','CSE',35,2,9.01),
(192511135,'Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:



The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' with columns: reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, and cgpa. The data is as follows:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhyia	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

Below the table, the 'Output' window shows the execution of SQL commands:

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

2. Create the Cursor Procedure

This cursor fetches the student_name values one by one and builds a list, just like the email-list experiment in your record.

```
DELIMITER $$
```

```
CREATE PROCEDURE build_student_list(INOUT student_list VARCHAR(4000))
BEGIN
```

```
    DECLARE v_finished INT DEFAULT 0;
    DECLARE v_student_name VARCHAR(100);
```

```
    DECLARE student_cursor CURSOR FOR
        SELECT student_name FROM student_cgpa;
```

```
    DECLARE CONTINUE HANDLER FOR NOT FOUND
        SET v_finished = 1;
```

```
    OPEN student_cursor;
```

```
    student_loop: LOOP
```

```
        FETCH student_cursor INTO v_student_name;
```

```
        IF v_finished = 1 THEN
            LEAVE student_loop;
        END IF;
```

```

        SET student_list = CONCAT(student_list, v_student_name, ' ');

    END LOOP student_loop;

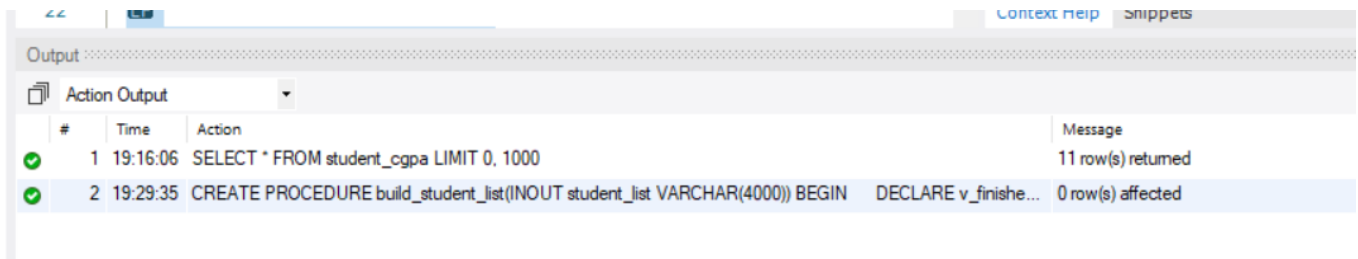
    CLOSE student_cursor;

END$$

DELIMITER ;

```

OUTPUT:



The screenshot shows the 'Output' window of a SQL IDE. It contains a table with columns: #, Time, Action, and Message. Two entries are visible, both marked with a green checkmark icon.

#	Time	Action	Message
1	19:16:06	SELECT * FROM student_cgpa LIMIT 0, 1000	11 row(s) returned
2	19:29:35	CREATE PROCEDURE build_student_list(INOUT student_list VARCHAR(4000)) BEGIN DECLARE v_finishe...	0 row(s) affected

3. EXECUTION

```

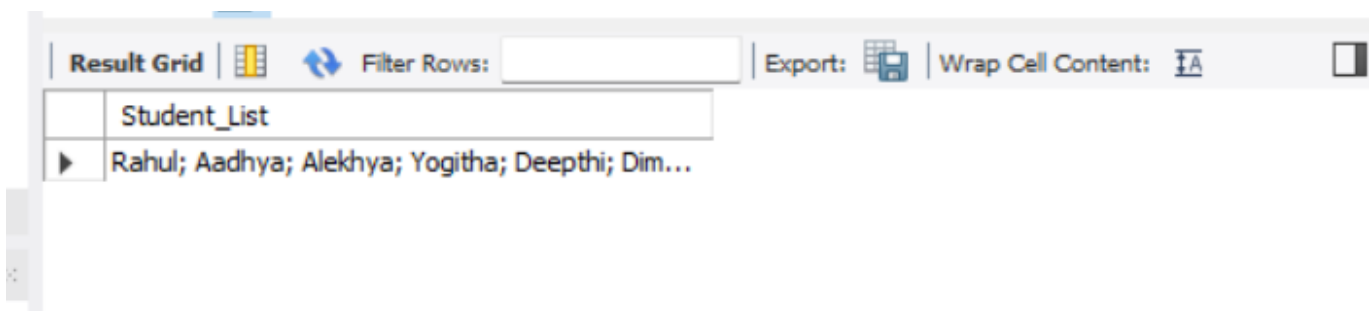
SET @student_list = "";

CALL build_student_list(@student_list);

SELECT @student_list AS Student_List;

```

OUTPUT:



The screenshot shows the 'Result Grid' of a SQL IDE. It displays a single column named 'Student_List' with one row of data. The data is a string containing names separated by semicolons.

Student_List
Rahul; Aadhya; Alekhya; Yogitha; Deepthi; Dim...

RESULT: Thus the Cursor program using MySQL is executed successfully.5